WiFi Monitor

How to use your desk

This desk shows your own Wi-Fi. You get the signal, a check to the router, a check out to the internet, download and upload speed, a clock, and a few names and addresses for that connection. There is no app and no account. Leave it plugged in.

The photos in the shop are example screens. The names and numbers in this book are examples too, so you can see what each word means. Your desk fills in your own connection.

Two sizes use the same words. The 4.3 inch screen is wide and short. The 4.0 inch screen is a little taller. The Wi-Fi keyboard is not the same on the two sizes, and the 4.0 inch asks you to touch four corners the first time. The pages below say which size they mean. If a line is not on the page you are looking at, it is on the other page. Nothing here is a list of the other devices in the house.

What you need

A 2.4 GHz Wi-Fi name and its password. This desk does not join a 5 GHz-only network. If your house has two names and one of them ends in 5G, use the other name.

Plug in the power cable. Leave it plugged in if you want the numbers to keep updating.

The password is typed on the glass, and the glass shows the password while you are on the Wi-Fi page. Anyone who can see that page can read it. When you leave that page, the password is not on the home screen.

First time you plug it in

The screen lights up.

If this is a 4.0 inch desk and it has never learned your touch, it says CORNER 1 OF 4 and TOUCH THE ARROW. That step is the next section. A 4.3 inch desk does not ask for corners.

If no network is saved, it opens the Wi-Fi page and looks for networks. Join one, using the keyboard section for your size.

If a network is already saved, the screen says linking. That can take up to about 15 seconds. Then it opens the home screen. If it could not join, the home screen says no link. Tap WIFI and check the name and the password.

4.0 inch only: the four corners

This happens once, the first time that screen is used, and again only if that touch setup was cleared.

The screen says CORNER 1 OF 4 and TOUCH THE ARROW. A white arrow sits in one corner. Touch the tip of the arrow, not the middle of the glass. It moves to the next corner. Do all four: top left, bottom

left, top right, bottom right.

If a touch was not firm enough, it asks for that corner again. When all four are accepted, it stores them and goes on to Wi-Fi. Later, RESET on the Wi-Fi page forgets the network. It does not forget these four corners.

The buttons along the bottom

HOME is the main page.
INFO opens the link page. The button says INFO. The title on that page says LINK.
GRAPH opens the three charts.
WIFI opens the keyboard.

The button for the page you are on is gold. The others are dark. These four stay on every page except while you are typing a network.

Home screen

The gold title is WIFI MONITOR.

The clock is at the top. It is hours and minutes, with AM or PM. It does not show seconds. The time comes from the internet after Wi-Fi connects. The place under the title is the location that came back with that connection, such as Los Angeles. The date on that same line is the local date for that place, like Sun Oct 11.

Until that location comes back, the clock says --:-- and the line says setting clock. That is the desk waiting, not a broken clock. Give it a minute on Wi-Fi. There is no list of Pacific, Mountain, Central, or Eastern to tap.

The four bars are the signal. More green bars means a stronger signal.

4 bars means the signal is stronger than -55 dBm.
3 bars means it is stronger than -67.
2 bars means it is stronger than -75.
1 bar means it is connected, but weak.
If it is not connected, the bars stay dark and the name says no link.

The name next to the bars is the network you joined. Under it, on a live link, is the signal in dBm and the channel. dBm is the same strength as the bars, written as a number. A number closer to zero is stronger. -50 is strong. -80 is weak. The channel is the Wi-Fi channel number, such as 6.

download and upload are speeds in megabits per second, with one decimal, such as 48.2. The desk downloads a small file and uploads a small amount of data, then turns that into the number. It is a short check, not a long lab test. Right after you join, both can say -- for a few seconds while the first check runs. After that, it checks again about every 3 minutes. If a check does not finish, that line stays -- until the next one works.

house is the time, in milliseconds, for a small check to reach your router. A small number means the router answered quickly. house -- means that check did not get an answer.

internet is the same kind of check, sent to an address out on the internet. internet -- means that check did not get an answer. A house number with no internet number usually means the Wi-Fi to the router is up and the path out of the house is not.

connections is a count of addresses on your own network that answered. It is not a list. It does not show names, and it does not show which phone or TV is which. Phones that are asleep often do not answer, so the count can be lower than the client list in the router. The router itself usually answers, so it is part of the count. The count stays -- until the first walk across the network finishes. That walk starts after the first speed check, and it takes a little while. Then the desk counts again every few minutes.

company is the internet company name that came back with your public address. If that name does not come back, it says company --.

router is the name the router gives out for itself. If the router does not answer that question, it says router --.

this is the address of this desk on your network, such as 192.168.1.42.

public is the address your house shows to the rest of the internet.

gateway is the address of the router on your network.

dns is the address of the name server the desk was given. On the 4.3 inch home screen it is on this page. On the 4.0 inch it is on the LINK page.

best and worst are the fastest and slowest successful internet checks since midnight, in milliseconds. drops is how many internet checks failed since midnight. When the date changes, best, worst, and drops start over. Until the first successful check, best says --.

INFO, the link page

Tap INFO. The title says LINK.

ssid is the network name.
bssid is the wireless address of the router you joined, written as six pairs.
channel is the Wi-Fi channel, and the dBm next to it is the signal again.
mac is the wireless address of this desk. That is this desk, not the router.
dns is the name server address.
company, router, and public are the same words as on the home screen.
On the 4.3 inch, gateway and this are repeated here. On the 4.0 inch, gateway and this stay on the home screen.
down and up are the same speeds as download and upload on the home screen, in megabits per second. -- on this page is the same failed check as -- on the home screen. It is not a second test.

GRAPH

Three charts. Each title includes the latest number.

SIGNAL is the dBm, such as -58 dBm.
DOWNLOAD is the download speed, such as 48.2 Mbps.
INTERNET is the internet check, such as 18 ms.

The line is the recent readings. The up and down of the box is fit to those readings, so a small change is tall enough to see. A level line means that reading is holding steady. A steady signal often looks level. That is normal.

The boxes are empty until the desk has a couple of readings. That is the first half minute or so. A download line stays empty while every speed check is still --. It does not draw a fake zero.

Wi-Fi page on the 4.3 inch

Tap WIFI. The desk looks for networks and shows up to two names at the top. If your network is not one of those two, tap the name box and type it. There is no second page of names on this size.

name is the network. pw is the password. A gold outline is the box you are typing into. Tap the other box to switch.

The keys are small letters. 123 changes the keys to numbers and these symbols: - _ . : @ # + = / ( ) ! ? '

Tap 123 again to go back to letters. This size does not have a capital-letter key. If the password uses a capital letter, this keyboard cannot type it. Change the router password to small letters, or use a 4.0 inch desk, which has a capital key.

DEL erases the last character in the box you are typing in. SPACE types a space.

JOIN saves the name and password and tries to connect. It tries for about 12 seconds. If it joins, you go back to the home screen and the clock starts setting. If it does not join, you stay on this page. Check the name, the password, and that you used the 2.4 GHz network, then tap JOIN again.

FORGET clears the name and the password on this desk and disconnects. The router is not changed.

BACK returns to the home screen without joining.

RESET clears the saved network on this desk and restarts it. It does not change the router, the phones, or anything else in the house.

Wi-Fi page on the 4.0 inch

Tap WIFI. Tap SCAN to look for networks. Two names show at a time. < and > move through the rest. Tap a name to put it in the SSID box. The cursor moves to the password.

SSID is the network. PASS is the password. A gold outline is the box you are typing into. Tap the other

box to switch.

The letter keys start as small letters. abc switches between small letters and capitals. The key reads abc while you are on small letters, and ABC while you are on capitals. !# switches to symbols. While symbols are showing, that key reads ABC and takes you back to letters.

DEL erases the last character. SPACE types a space. A number row is always on the keyboard.

JOIN, FORGET, and RESET do the same jobs as on the 4.3 inch. BACK is at the top left on this size, and it returns home without joining.

RESET on this size forgets the saved network and restarts. The four-corner touch setup stays.

A few things that are normal

-- means that item has not arrived yet, or the last check failed. It is not a zero.

setting clock means the place has not come back yet. The time is not stuck on a time zone you picked. You do not pick one.

The first speed numbers can take a few seconds. The connection count can take longer than the speeds, because it walks the addresses in the house after that first speed check.

A level line on the graph means the reading is steady.

router -- means the router did not give out a name. The other lines can still be fine.

company -- means the company name did not come back. The speed and the signal do not depend on it.

no link means it is not joined. Tap WIFI.

If the Wi-Fi drops later, the desk keeps the last numbers it has until the next check. The bars go dark when the link is gone.

What this desk does not do

It does not list the other devices. connections is only a count of addresses that answered.

It does not show who those addresses are.

It does not open the router's own settings page.

It does not join 5 GHz-only Wi-Fi.

It does not keep time while the power is unplugged. Plug it back in, let it join, and the clock sets again from the internet.